

MA1001 — Calculus for Science Part 1

Winter 2025 Examination · Paper 1 · Full Marking Scheme

Module	MA1001 — Calculus for Science Part 1
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Institution	University College Cork
Session	Winter 2025, Paper 1
Duration	1.5 hours (+ 15 minutes reading time)
Structure	Answer all 6 questions . Each question is worth 10 points (60 in total). Marks for parts are shown in square brackets.
Materials	Formula sheet supplied with the paper (calculus, trigonometry, length/area/volume)

How to use this scheme. Every answer is worked out in small numbered steps, with a “Rules you need” box listing the rules used. Nothing is skipped, so you can see where every number comes from. Read the answer first, then follow the steps.

Coverage note. This scheme works through **all six questions and every part** in full. There are no either/or options on this paper.

Source basis. Every answer is built **only** from the MA1001 lecture notes and example/tutorial sheets: logarithms, graph sketching and transformations, derivative rules, increasing/decreasing functions and the first derivative test, asymptotes, and rates of change. No textbook or outside material is used.

Mark splits. The paper only splits marks for Question 1. For Questions 2–6 (10 points each), the split shown is Study-Uni's suggested allocation based on the key steps.

Own-figure rule. Where a later step uses an earlier result, a correct method applied to a wrong earlier value still earns the method marks.

Numbers. Exact answers ($\ln 2$, $\sin 1$) are given first, with decimals to 3 significant figures. Keep your calculator in radian mode throughout.

Study-Uni

Question 1

Piecewise functions, logarithms, graph transformations — 10 marks

Given: $f(x) = \ln(1 - x)$ if $x < 0$; 0 if $0 \leq x < 1$; $\ln(1 + x)$ if $x \geq 1$

1(a)

[2 marks]

What are the values $f(0)$ and $f(1)$?

ANSWER

$f(0) = 0$ and $f(1) = \ln 2 \approx 0.693$

WORKING — STEP BY STEP

1 How to read a piecewise function

f has three different rules. For any number x , you check which condition x satisfies, and use **only** that rule.

Rule 1: $x < 0$ | Rule 2: $0 \leq x < 1$ | Rule 3: $x \geq 1$

2 Find $f(0)$: which rule does $x = 0$ belong to?

Rule 1 needs $0 < 0$. That's false (0 is not less than itself).

Rule 2 needs $0 \leq 0$ **and** $0 < 1$. Both are true, so we use Rule 2.

Rule 2 just says the answer is 0, whatever x is:

$$f(0) = 0$$

3 Find $f(1)$: which rule does $x = 1$ belong to?

Rule 1 needs $1 < 0$: false. Rule 2 needs $1 < 1$: false (1 is not less than 1).

Rule 3 needs $1 \geq 1$: true (1 is equal to 1, and $\geq$ allows equal). So use Rule 3, $\ln(1 + x)$, with $x = 1$:

$$f(1) = \ln(1 + 1) = \ln 2$$

On the calculator press **ln 2**:

$$f(1) = \ln 2 \approx 0.693$$

EXAM TECHNIQUE

The end points are the trap. The sign $\leq$ or $\geq$ includes the number itself; the sign $<$ or $>$ does not. So 0 goes with $0 \leq x < 1$, and 1 goes with $x \geq 1$.

Marks

Awarded for

1 $f(0) = 0$ (correct rule chosen).

1 $f(1) = \ln 2$ (≈ 0.693).

1(b)**[2 marks]**Is f decreasing on $(-\infty, -1)$? Justify your answer.**ANSWER**

Yes. On $(-\infty, -1)$, $f'(x) = -1/(1 - x)$, which is negative, so f is decreasing.

RULES YOU NEED

- f is decreasing where $f'(x) < 0$.
- Derivative of $\ln u$ is $1/u$.
- Chain rule: $\frac{df}{dx} = \frac{df}{du} \times \frac{du}{dx}$.

WORKING — STEP BY STEP**1 Which rule of f do we use here?**

$(-\infty, -1)$ means every number less than -1 (e.g. $-2, -5, -100$). All of these are less than 0 , so Rule 1 applies:

$$f(x) = \ln(1 - x)$$

2 The key idea: the sign of the derivative

A function is **decreasing** where its derivative is **negative**, $f'(x) < 0$, and increasing where $f'(x) > 0$. So we differentiate and check the sign.

3 Differentiate $\ln(1 - x)$ using the chain rule

There is a function inside a function: $\ln()$ on the outside, $1 - x$ on the inside. Let $u = 1 - x$.

$$f = \ln u \Rightarrow \frac{df}{du} = \frac{1}{u} \quad u = 1 - x \Rightarrow \frac{du}{dx} = -1$$

Chain rule: multiply the two derivatives, then put $u = 1 - x$ back in:

$$f'(x) = \frac{1}{1 - x} \times (-1) = -\frac{1}{1 - x}$$

4 Is $f'(x)$ negative when $x < -1$?

If x is less than -1 , then $1 - x$ is bigger than 2 . For example, $x = -3$ gives $1 - (-3) = 1 + 3 = 4$.

So $1/(1 - x)$ is a positive number, and the minus sign in front makes $f'(x)$ negative:

$$x = -3: f'(-3) = -1/4 = -0.25 \quad (\text{negative})$$

$$f'(x) < 0 \text{ for every } x \text{ in } (-\infty, -1)$$

5 Conclusion (and a quick check with numbers)

Since $f'(x) < 0$ on the whole interval, **yes, f is decreasing on $(-\infty, -1)$.**

Check: as x goes up from -3 to -2 , f goes down:

$$f(-3) = \ln 4 \approx 1.386 \rightarrow f(-2) = \ln 3 \approx 1.099$$

Marks	Awarded for
1	$f'(x) = -1/(1 - x)$ using the chain rule (or a correct argument that $\ln(1 - x)$ gets smaller as x increases).
1	$f'(x) < 0$ on $(-\infty, -1)$, so yes, f is decreasing.

1(c)**[3 marks]**Sketch the graph of $y = f(x)$ for $-2 < x < 2$.**ANSWER**

See the sketch: a falling curve from $(-2, \ln 3)$ down to $(0, 0)$, flat along the x-axis up to $x = 1$, then a jump up to $(1, \ln 2)$ and a rising curve up to $(2, \ln 3)$.

WORKING — STEP BY STEP**1 Split the range $-2 < x < 2$ into the three rules**

$$-2 < x < 0: y = \ln(1 - x) \quad | \quad 0 \leq x < 1: y = 0 \quad | \quad 1 \leq x < 2: y = \ln(1 + x)$$

2 Left piece: make a small table of values for $y = \ln(1 - x)$ Put a few x values into $\ln(1 - x)$ with the calculator:

x	-2	-1.5	-1	-0.5	0
1 - x	3	2.5	2	1.5	1
y = $\ln(1 - x)$	1.10	0.92	0.69	0.41	0

The y values go **down** from about 1.10 to 0, matching part (b). The curve meets the origin $(0, 0)$.

3 Middle piece: $y = 0$ for $0 \leq x < 1$ The graph sits right on the x -axis from $x = 0$ to $x = 1$.At $x = 0$: $\leq$ means 0 is included, so draw a **filled dot** at $(0, 0)$.At $x = 1$: $<$ means 1 is **not** included, so draw an **open dot** at $(1, 0)$.**4 Right piece: table of values for $y = \ln(1 + x)$**

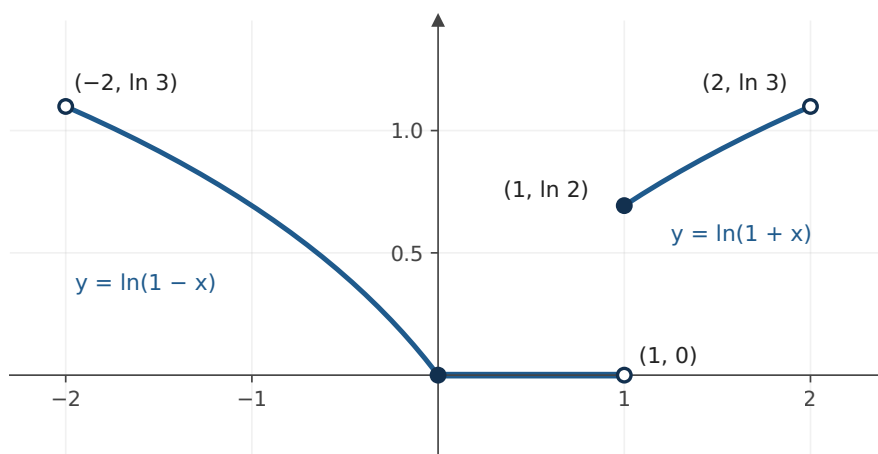
x	1	1.5	2
1 + x	2	2.5	3
y = $\ln(1 + x)$	0.69	0.92	1.10

At $x = 1$: $\geq$ means 1 is included, so draw a **filled dot** at $(1, \ln 2) \approx (1, 0.69)$. The curve rises from there.

So at $x = 1$ the graph **jumps** from the open dot at 0 up to the filled dot at 0.69.

5 The two ends

The question says $-2 < x < 2$, so $x = -2$ and $x = 2$ are not included: draw **open dots** at $(-2, \ln 3)$ and $(2, \ln 3)$, both at height ≈ 1.10 .



Marks	Awarded for
1	Falling curve from about $(-2, 1.10)$ down to the origin.
1	Flat line on the x-axis for $0 \leq x < 1$, with an open dot at $(1, 0)$.
1	Jump to a filled dot at $(1, \ln 2)$, then a rising curve towards $(2, \ln 3)$.

1(d)**[3 marks]**Sketch the graph of $g(x) = f(-x) + 1$ for $-2 < x < 2$.**ANSWER**

$g(x) = 1 + \ln(1 - x)$ for $x \leq -1$; 1 for $-1 < x \leq 0$; $1 + \ln(1 + x)$ for $x > 0$ (see sketch).

WORKING — STEP BY STEP**1 What the two changes do to a graph**

$f(-x)$: put $-x$ in place of x . This flips the graph left-to-right (a mirror image in the y -axis).

$+ 1$: adds 1 to every y value. This moves the whole graph up by 1.

2 Work out $f(-x)$: replace x by $-x$ in each rule

Each condition changes too. Multiplying an inequality by -1 **flips its direction** (e.g. $-x < 0$ becomes $x > 0$).

$$\text{Rule 1: } -x < 0 \Leftrightarrow x > 0 \text{ gives } \ln(1 - (-x)) = \ln(1 + x)$$

$$\text{Rule 2: } 0 \leq -x < 1 \Leftrightarrow -1 < x \leq 0 \text{ gives } 0$$

$$\text{Rule 3: } -x \geq 1 \Leftrightarrow x \leq -1 \text{ gives } \ln(1 + (-x)) = \ln(1 - x)$$

3 Add 1 to each rule to get $g(x)$

$$g(x) = 1 + \ln(1 - x) \text{ if } x \leq -1$$

$$g(x) = 1 \text{ if } -1 < x \leq 0$$

$$g(x) = 1 + \ln(1 + x) \text{ if } x > 0$$

4 Table of values

x	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2
$g(x)$	2.10	1.92	1.69	1	1	1.41	1.69	1.92	2.10

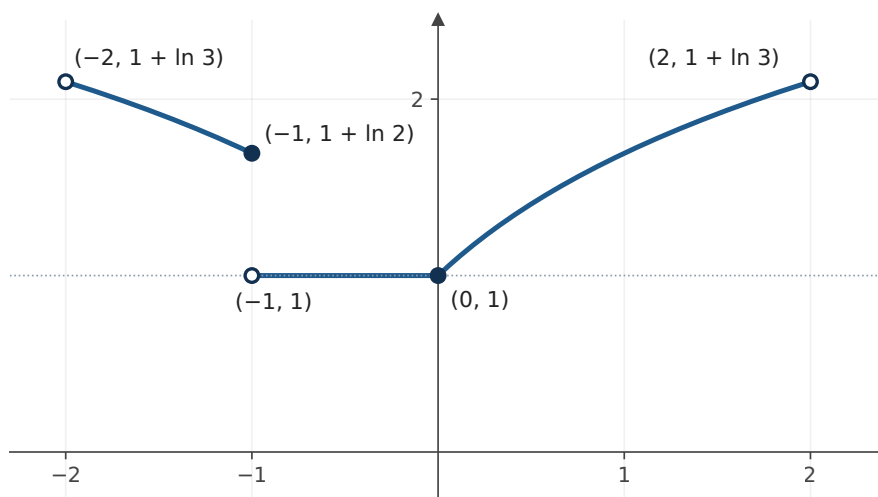
For example, $g(-2) = 1 + \ln(1 - (-2)) = 1 + \ln 3 = 1 + 1.10 = 2.10$, and $g(0.5) = 1 + \ln 1.5 = 1 + 0.41 = 1.41$.

5 Dots at the break points

At $x = -1$: the first rule has $\leq$, so a **filled dot** at $(-1, 1 + \ln 2) \approx (-1, 1.69)$. The middle rule has $<$, so an **open dot** at $(-1, 1)$. The graph jumps down here.

At $x = 0$: the middle rule includes 0 ($\leq$), so a **filled dot** at $(0, 1)$. The right curve starts from there and rises.

The ends $x = \pm 2$ are not included, so draw **open dots** at height ≈ 2.10 .



EXAM TECHNIQUE

Do the mirror flip first, then move up by 1. The jump that was at $x = 1$ in part (c) is now at $x = -1$.

Marks	Awarded for
1	$f(-x)$ worked out correctly (mirror image in the y-axis).
1	Shifted up by 1: left curve ending at a filled dot $(-1, 1 + \ln 2)$, then a jump down to an open dot $(-1, 1)$.
1	Flat at $y = 1$ on $(-1, 0]$, then a rising curve towards $(2, 1 + \ln 3)$.

Question 1 Total: 10 Marks

Question 2

Logarithm rules, solving equations — 10 marks

2

[10 marks]

Solve the equation $\log_2(x^2 - 6x) = 3 + \log_2(1 - x)$.

ANSWER

$x = -4$ ($x = 2$ also comes out of the algebra but must be rejected)

RULES YOU NEED

- $\log A - \log B = \log(A \div B)$
- $\log_2 y = 3$ means $y = 2^3 = 8$
- You can only take the log of a positive number.

WORKING — STEP BY STEP

1 Find which x values are allowed

You can only take the log of a **positive** number. So both inputs must be positive:

$$x^2 - 6x > 0 \quad \text{and} \quad 1 - x > 0$$

$1 - x > 0$ means $x < 1$.

For $x^2 - 6x$, factor out x : $x^2 - 6x = x(x - 6)$. Test a number in each region:

$$x = -1: (-1)(-7) = 7 \checkmark \quad x = 1: (1)(-5) = -5 \times \quad x = 7: (7)(1) = 7 \checkmark$$

So $x^2 - 6x > 0$ when $x < 0$ or $x > 6$. Combined with $x < 1$, only this is allowed:

$$x < 0$$

2 Get both logs on the same side

Subtract $\log_2(1 - x)$ from both sides:

$$\log_2(x^2 - 6x) - \log_2(1 - x) = 3$$

3 Combine the two logs into one

Use the rule: $\log A - \log B = \log(A \div B)$.

$$\log_2 \frac{x^2 - 6x}{1 - x} = 3$$

4 Remove the log

$\log_2(\text{something}) = 3$ means the something equals 2^3 , and $2^3 = 2 \times 2 \times 2 = 8$:

$$\frac{x^2 - 6x}{1 - x} = 8$$

5 Get rid of the fraction

Multiply both sides by $(1 - x)$, then multiply out the bracket:

$$x^2 - 6x = 8(1 - x) = 8 - 8x$$

6 Rearrange into a quadratic (everything on one side, = 0)

Add $8x$ to both sides and subtract 8 from both sides:

$$x^2 - 6x + 8x - 8 = 0 \Rightarrow x^2 + 2x - 8 = 0$$

7 Solve the quadratic

Find two numbers that **multiply to -8** and **add to +2**: these are $+4$ and -2 ($4 \times -2 = -8$ and $4 + (-2) = 2$).

$$(x + 4)(x - 2) = 0 \Rightarrow x = -4 \text{ or } x = 2$$

(The formula $x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$ gives the same: $(-2 \pm \sqrt{36})/2 = (-2 \pm 6)/2 = 2$ or -4 .)

8 Check both answers against Step 1

$x = 2$: $x^2 - 6x = 4 - 12 = -8$, which is negative, so $\log_2(-8)$ doesn't exist. **Reject $x = 2$.**

$x = -4$: $x^2 - 6x = 16 + 24 = 40$ and $1 - x = 1 + 4 = 5$, both positive ✓. Check it in the original equation:

$$\text{Left: } \log_2 40 \approx 5.32 \quad \text{Right: } 3 + \log_2 5 \approx 3 + 2.32 = 5.32 \quad \checkmark$$

(On a calculator, $\log_2 40 = \ln 40 \div \ln 2$.) So the only solution is **$x = -4$** .

EXAM TECHNIQUE

Always check every answer in the original equation. Combining the logs hides the rule that each log needs a positive input, so a fake answer can appear (here $x = 2$).

Marks	Awarded for
2	Conditions $x^2 - 6x > 0$ and $1 - x > 0$ stated (or both roots checked in the original equation).
2	Logs combined using $\log A - \log B = \log(A/B)$ (or writing $3 = \log_2 8$).
1	$(x^2 - 6x)/(1 - x) = 8$.
2	Quadratic $x^2 + 2x - 8 = 0$.
1	Roots $x = -4$ and $x = 2$.
2	$x = 2$ rejected with a reason; final answer $x = -4$.

Question 2 Total: 10 Marks

Question 3

Chain rule, tangent lines — 10 marks

3

[10 marks]

Find the equation of the tangent line to the curve $y = \sin(\cos(x))$ at the point where $x = 0$.

ANSWER

$y = \sin 1 \quad (\approx 0.841)$: a flat, horizontal line

RULES YOU NEED

- Tangent line: $y - y_1 = m(x - x_1)$, where $m = dy/dx$ at the point.
- Chain rule: $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$.
- Derivatives: $\sin \rightarrow \cos$, and $\cos \rightarrow -\sin$. Trig table: $\cos 0 = 1$, $\sin 0 = 0$.

WORKING — STEP BY STEP

1 What we need for a tangent line

A straight line needs **a point** (x_1, y_1) and **a slope** m . Then use:

$$y - y_1 = m(x - x_1)$$

The slope of the tangent is the derivative dy/dx at that point.

2 Find the point (put $x = 0$ into the curve)

Work from the inside out. From the trig table, $\cos 0 = 1$:

$$y_1 = \sin(\cos 0) = \sin(1)$$

Put your calculator in **radians** and press $\sin 1$:

$$\text{Point: } (0, \sin 1) \approx (0, 0.841)$$

3 Differentiate using the chain rule

$y = \sin(\cos x)$ is a function inside a function: $\sin()$ on the outside, $\cos x$ on the inside. Let $u = \cos x$.

$$y = \sin u \Rightarrow \frac{dy}{du} = \cos u \quad u = \cos x \Rightarrow \frac{du}{dx} = -\sin x$$

Multiply them together, then put $u = \cos x$ back in:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \cos(\cos x) \times (-\sin x)$$

4 Find the slope at $x = 0$

From the trig table, $\sin 0 = 0$. Anything multiplied by 0 is 0:

$$m = \cos(\cos 0) \times (-\sin 0) = \cos(1) \times 0 = 0$$

A slope of 0 means the tangent line is flat (horizontal).

5 Write down the equation of the line

Put $x_1 = 0$, $y_1 = \sin 1$ and $m = 0$ into $y - y_1 = m(x - x_1)$:

$$y - \sin 1 = 0 \times (x - 0) = 0$$

$$y = \sin 1 \approx 0.841$$

EXAM TECHNIQUE

Keep the calculator in **radians**: $\sin 1 \approx 0.841$. In degree mode you'd get 0.0175, which is wrong. Leaving the answer as $y = \sin 1$ is also fine.

Marks	Awarded for
2	Point (0, sin 1).
3	$dy/dx = -\sin x \cdot \cos(\cos x)$ using the chain rule.
2	Slope $m = 0$ at $x = 0$.
3	Tangent line $y = \sin 1$ (≈ 0.841).

Question 3 Total: 10 Marks

Question 4

Related rates of change — 10 marks

4

[10 marks]

A car is driving along a straight road toward a streetlight that is 4.5 m tall. The car is 1.8 m tall and is moving at a speed of 9.0 m/s. As the car approaches the streetlight, it casts a shadow on the road. How fast is the length of the shadow changing when the car is 12 m away from the base of the streetlight?

ANSWER

$ds/dt = -6.0$ m/s: the shadow is getting shorter at 6.0 m/s

RULES YOU NEED

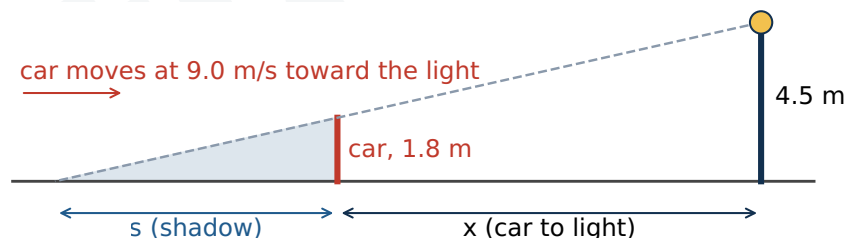
- Similar triangles: height $\div$ base is the same in both.
- Related rates: write an equation linking the lengths, then differentiate both sides with respect to time t .

WORKING — STEP BY STEP

1 Draw a picture and name the lengths

Treat the car as a straight upright line 1.8 m tall (like a person standing). Light from the top of the lamp passes the top of the car and hits the road at the tip of the shadow.

Let x = distance from the car to the base of the lamp, and s = length of the shadow (see the diagram).



2 Write down what we know and what we want

The car moves toward the lamp at 9.0 m/s, so x is getting **smaller**. That makes its rate negative:

$$\frac{dx}{dt} = -9.0 \text{ m/s}$$

We want $\frac{ds}{dt}$ (how fast the shadow length changes) when $x = 12$ m.

3 Link s and x using similar triangles

There are two right-angled triangles with the same shape:

Big triangle: height 4.5 m (lamp), base $x + s$ (from the lamp to the tip of the shadow).

Small triangle: height 1.8 m (car), base s (the shadow).

Same shape means height $\div$ base is the same for both:

$$\frac{4.5}{x + s} = \frac{1.8}{s}$$

4 Rearrange to get s on its own

Cross-multiply (multiply both sides by s and by $(x + s)$):

$$4.5 \times s = 1.8 \times (x + s) \Rightarrow 4.5s = 1.8x + 1.8s$$

Subtract $1.8s$ from both sides ($4.5 - 1.8 = 2.7$):

$$2.7s = 1.8x$$

Divide both sides by 2.7. Note $1.8 \div 2.7 = 18 \div 27 = 2/3$:

$$s = \frac{2}{3} x$$

5 Differentiate both sides with respect to time t

The number $2/3$ just stays in front:

$$\frac{ds}{dt} = \frac{2}{3} \times \frac{dx}{dt}$$

6 Put in the numbers

$$\frac{ds}{dt} = \frac{2}{3} \times (-9.0) = \frac{-18}{3} = -6.0 \text{ m/s}$$

The minus sign means the shadow is getting **shorter**, at 6.0 m/s.

At $x = 12$ m the shadow is $s = 2/3 \times 12 = 8$ m long. The rate doesn't depend on x (s is always $2/3$ of x), so it is -6.0 m/s at 12 m and at every other distance too.

EXAM TECHNIQUE

Getting closer means the distance is going down, so its rate is negative. A negative answer for ds/dt means the shadow is shrinking.

Marks	Awarded for
2	Diagram with variables, and $dx/dt = -9$ (negative because the car is approaching).
3	Similar-triangles equation $4.5/(x + s) = 1.8/s$.
1	$s = (2/3)x$.
2	Differentiating with respect to t : $ds/dt = (2/3) dx/dt$.
2	$ds/dt = -6.0$ m/s, i.e. the shadow is shrinking at 6 m/s.

Question 4 Total: 10 Marks

Question 5

Critical points, increasing and decreasing functions, curve sketching — 10 marks

Given: $f(x) = \frac{x^2}{x^4 + 1}$

5 · Critical points

[4 marks]

Consider the function $f(x) = \frac{x^2}{x^4 + 1}$ defined for any real number x . Find all of the critical points of this function.

ANSWER

Critical points at $x = -1, 0$ and 1 , i.e. the points $(-1, \frac{1}{2})$, $(0, 0)$ and $(1, \frac{1}{2})$

RULES YOU NEED

- Critical point: where $f'(x) = 0$.
- Quotient rule: if $f = u/v$ then $f' = (u' \cdot v - u \cdot v') / v^2$.
- Derivative of x^n is $n \cdot x^{n-1}$.
- Tidy the top, but never multiply out the bottom.

WORKING — STEP BY STEP

1 What a critical point is

A critical point is where the slope is zero: $f'(x) = 0$. So we differentiate, then solve $f'(x) = 0$.

2 Set up the quotient rule

f is a fraction, so use the quotient rule (on the formula sheet). Name the top u and the bottom v , and differentiate each:

$$u = x^2 \Rightarrow u' = 2x \quad v = x^4 + 1 \Rightarrow v' = 4x^3$$

$$f'(x) = \frac{u' \cdot v - u \cdot v'}{v^2}$$

3 Substitute into the formula

$$f'(x) = \frac{(2x)(x^4 + 1) - (x^2)(4x^3)}{(x^4 + 1)^2}$$

4 Tidy up the top (never multiply out the bottom)

Multiply out each part. When you multiply powers, add the indices ($x^2 \times x^3 = x^5$):

$$(2x)(x^4 + 1) = 2x^5 + 2x \quad (x^2)(4x^3) = 4x^5$$

$$\text{Top} = 2x^5 + 2x - 4x^5 = 2x - 2x^5$$

Take out the common factor $2x$, then use the difference of squares twice ($a^2 - b^2 = (a - b)(a + b)$):

$$2x - 2x^5 = 2x(1 - x^4) = 2x(1 - x^2)(1 + x^2) = 2x(1 - x)(1 + x)(1 + x^2)$$

$$f'(x) = \frac{2x(1 - x)(1 + x)(1 + x^2)}{(x^4 + 1)^2}$$

5 Solve $f'(x) = 0$

The bottom $(x^4 + 1)^2$ is always positive, never 0. A fraction is 0 only when its top is 0, so set each factor to 0:

$$2x = 0 \Rightarrow x = 0 \quad 1 - x = 0 \Rightarrow x = 1 \quad 1 + x = 0 \Rightarrow x = -1$$

$1 + x^2 = 0$ has no solution (x^2 can't be negative), so it gives nothing.

6 Find the y-value at each critical point

Put each x back into the original $f(x)$:

$$f(0) = \frac{0}{0 + 1} = 0 \quad f(1) = \frac{1}{1 + 1} = \frac{1}{2} \quad f(-1) = \frac{(-1)^2}{(-1)^4 + 1} = \frac{1}{2} = \frac{1}{2}$$

Marks	Awarded for
2	$f'(x)$ found using the quotient rule.
1	Top simplified to $2x(1 - x^4)$ and set equal to 0.
1	$x = -1, 0, 1$ with their y-values.

5 · Increasing / decreasing**[3 marks]**

Find the intervals on which the function is increasing, and those on which it is decreasing.

ANSWER

Increasing on $(-\infty, -1)$ and $(0, 1)$; decreasing on $(-1, 0)$ and $(1, \infty)$. Local maximums at $x = \pm 1$, local minimum at $x = 0$.

WORKING — STEP BY STEP**1 The rule**

f is **increasing** where $f'(x) > 0$ and **decreasing** where $f'(x) < 0$. The bottom of f' is always positive, so only the top, $2x(1 - x^4)$, decides the sign.

2 Mark the split points and choose test points

The **split points** are the critical points $-1, 0$ and 1 . They cut the number line into four pieces, and f' can only change sign at them. Pick one **test point** in each piece:

$(-\infty, -1)$: test $x = -2$ $(-1, 0)$: test $x = -0.5$ $(0, 1)$: test $x = 0.5$ $(1, \infty)$: test $x = 2$

3 Work out the sign of $2x(1 - x^4)$ at each test point

$x = -2$: $2x = -4$, $1 - x^4 = 1 - 16 = -15$, product = $(-4)(-15) = +60 \rightarrow$ positive

$x = -0.5$: $2x = -1$, $1 - x^4 = 1 - 0.0625 = 0.9375$, product = $-0.9375 \rightarrow$ negative

$x = 0.5$: $2x = 1$, $1 - x^4 = 0.9375$, product = $+0.9375 \rightarrow$ positive

$x = 2$: $2x = 4$, $1 - x^4 = -15$, product = $-60 \rightarrow$ negative

4 Summary table

interval	$(-\infty, -1)$	$(-1, 0)$	$(0, 1)$	$(1, \infty)$
sign of $f'(x)$	+	−	+	−
f is	increasing ↗	decreasing ↘	increasing ↗	decreasing ↘

5 Classify each critical point (first derivative test)

At $x = -1$ the graph goes up then down: a **hill**, so a local maximum, value $\frac{1}{2}$.

At $x = 0$ it goes down then up: a **valley**, so a local minimum, value 0 .

At $x = 1$ it goes up then down: a local maximum, value $\frac{1}{2}$.

Marks**Awarded for****2**Correct sign of f' on each of the four intervals.**1**

Intervals stated and critical points classified.

5 · Sketch**[3 marks]**

Use all of this information to give a sketch of the graph of the function.

ANSWER

Two equal hills at $(\pm 1, \frac{1}{2})$, a valley touching the origin, and both ends flattening out towards $y = 0$ (see the sketch).

WORKING — STEP BY STEP**1 Plot the critical points**

Hills (local maximums) at $(-1, \frac{1}{2})$ and $(1, \frac{1}{2})$. Valley (local minimum) at $(0, 0)$.

2 Check for symmetry

Put $-x$ into f . Even powers remove the minus sign: $(-x)^2 = x^2$ and $(-x)^4 = x^4$.

$$f(-x) = \frac{(-x)^2}{(-x)^4 + 1} = \frac{x^2}{x^4 + 1} = f(x)$$

So the graph is a mirror image of itself in the y -axis (the left side mirrors the right side).

3 Is the graph ever below the x -axis?

The top x^2 is never negative and the bottom $x^4 + 1$ is always positive, so $f(x) \geq 0$. The graph never goes below the x -axis.

4 What happens far out (x very large)?

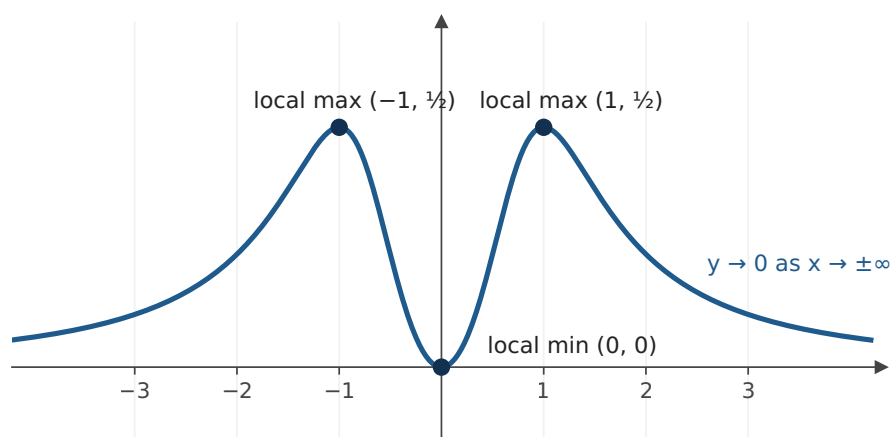
Divide the top and bottom by x^4 (the biggest power):

$$f(x) = \frac{x^2 / x^4}{x^4/x^4 + 1/x^4} = \frac{1/x^2}{1 + 1/x^4} \rightarrow \frac{0}{1 + 0} = 0$$

So $y = 0$ (the x -axis) is a **horizontal asymptote**: the graph gets closer and closer to it. Check: $f(10) = 100 \div 10\,001 \approx 0.01$, very close to 0.

5 Join it all up

Coming in from the left close to the x -axis, rise to the hill at $(-1, \frac{1}{2})$, drop to the valley at $(0, 0)$, rise to the hill at $(1, \frac{1}{2})$, then fall back towards the x -axis. This matches the increasing/decreasing table.



Marks	Awarded for
1	Symmetry, $f \geq 0$, or the horizontal asymptote $y = 0$ used.
2	Sketch with maximums at $(\pm 1, \frac{1}{2})$, minimum at $(0, 0)$, rising and falling in the right places.

Question 5 Total: 10 Marks

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Question 6

Checking solutions of a differential equation — 10 marks

Given: $\frac{df}{dx} = -f + e^{-x}$. Overall answer: only (c), $f(x) = xe^{-x}$, satisfies it.

6(a)

[2 marks]

Does $f(x) = \cos x$ satisfy $\frac{df}{dx} = -f + e^{-x}$?

ANSWER

No

RULES YOU NEED

- Work out the **left side**, df/dx (the derivative of f).
- Work out the **right side**, $-f + e^{-x}$ (put f in, with a minus in front, and add e^{-x}).
- Same for **every** $x \rightarrow$ yes. Different for even **one** $x \rightarrow$ no.

WORKING — STEP BY STEP

1 Left side: differentiate $\cos x$

$$\frac{df}{dx} = -\sin x$$

2 Right side: put $f = \cos x$ in

$$-f + e^{-x} = -\cos x + e^{-x}$$

3 Compare using one value of x

Try $x = \pi/2$, where $\sin(\pi/2) = 1$ and $\cos(\pi/2) = 0$ (trig table):

$$\text{Left: } -\sin(\pi/2) = -1 \quad \text{Right: } -0 + e^{-\pi/2} \approx 0.208$$

-1 is not equal to 0.208 , so $\cos x$ does **not** satisfy the equation.

Marks

Awarded for

1 $df/dx = -\sin x$ and the right side $-\cos x + e^{-x}$.

1 Shown not equal (e.g. at $x = \pi/2$): No.

6(b)**[2 marks]**

Does $f(x) = e^{-x}$ satisfy $\frac{df}{dx} = -f + e^{-x}$?

ANSWER**No****WORKING — STEP BY STEP****1 Left side: differentiate e^{-x}**

The formula sheet says the derivative of e^{ax} is $a \cdot e^{ax}$. Here $a = -1$:

$$\frac{df}{dx} = -1 \times e^{-x} = -e^{-x}$$

2 Right side: put $f = e^{-x}$ in

$$-f + e^{-x} = -e^{-x} + e^{-x} = 0$$

3 Compare

The left side is $-e^{-x}$. e to any power is always positive, so $-e^{-x}$ is never 0. For example, at $x = 0$: left = -1 , right = 0.

They are never equal, so the answer is **no**.

Marks**Awarded for****1** $df/dx = -e^{-x}$ and the right side = 0.**1** $-e^{-x} \neq 0$, so No.

6(c)**[4 marks]**

Does $f(x) = xe^{-x}$ satisfy $\frac{df}{dx} = -f + e^{-x}$?

ANSWER

Yes: this is the function that satisfies the equation

RULES YOU NEED

- Product rule: if $f = u \cdot v$ then $f' = u \cdot v' + v \cdot u'$.
- Derivative of e^{-x} is $-e^{-x}$.

WORKING — STEP BY STEP**1 Left side: use the product rule**

$f = x \times e^{-x}$ is two things multiplied, so use the product rule. Name them and differentiate each:

$$u = x \Rightarrow u' = 1 \quad v = e^{-x} \Rightarrow v' = -e^{-x}$$

$$\frac{df}{dx} = u \cdot v' + v \cdot u' = x \times (-e^{-x}) + e^{-x} \times 1 = e^{-x} - xe^{-x}$$

2 Right side: put $f = xe^{-x}$ in

$$-f + e^{-x} = -xe^{-x} + e^{-x}$$

3 Compare

$$\text{Left: } e^{-x} - xe^{-x} \quad \text{Right: } -xe^{-x} + e^{-x}$$

These are the same two terms, just written in a different order. So they are equal for **every** x .

Quick check at $x = 1$: left = $e^{-1} - e^{-1} = 0$ and right = $-e^{-1} + e^{-1} = 0 \checkmark$

So **yes**, $f(x) = xe^{-x}$ satisfies the equation.

Marks	Awarded for
2	$df/dx = e^{-x} - xe^{-x}$ using the product rule.
1	Right side $-xe^{-x} + e^{-x}$.
1	Shown equal for all x : Yes.

6(d)**[2 marks]**Does $f(x) = x$ satisfy $\frac{df}{dx} = -f + e^{-x}$?**ANSWER****No****WORKING — STEP BY STEP****1 Left side: differentiate x**

$$\frac{df}{dx} = 1$$

2 Right side: put $f = x$ in

$$-f + e^{-x} = -x + e^{-x}$$

3 Compare using one value of xTry $x = 1$ ($e^{-1} \approx 0.368$ on the calculator):

$$\text{Left: } 1 \quad \text{Right: } -1 + 0.368 = -0.632$$

1 is not equal to -0.632 , so the answer is **no**.**EXAM TECHNIQUE**

Choose your test value carefully. At $x = 0$ both sides of (d) equal 1, so testing only $x = 0$ would wrongly suggest yes. To say “yes” you must show the two sides match for every x , as in (c).

Marks**Awarded for****1** $df/dx = 1$ and the right side $-x + e^{-x}$.**1** Shown not equal (e.g. at $x = 1$): No.**Question 6 Total: 10 Marks**